

Advanced Econometrics

Models for count data

Ewa Weychert, PhD

What is Count Data?

- Data that counts things
- Only takes values: 0,1,2,3,...
- Cannot be negative or decimal

Examples:

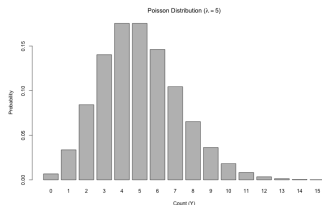
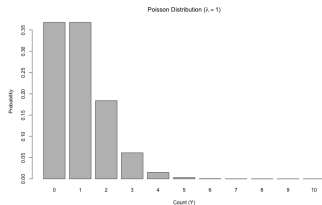
- Number of accidents
- Number of hospital visits
- Number of emails received

Why Not Use Linear Regression?

- Linear regression can predict negative values
- Count data is always non-negative
- Variance is often not constant

So we need special models!

Poisson Distribution



- Used for counting things: $0, 1, 2, \dots$
- One key number: λ (average count)

Model: $[Y \sim \text{Poisson}(\lambda)]$

- Y = actual count
- λ = expected count

Regression version: $[\log(\lambda) = X\beta]$

- X = inputs
- β = importance of inputs
- log link keeps prediction positive

$$[\lambda = e^{X\beta}]$$

Small $\lambda \rightarrow$ many zeros Bigger $\lambda \rightarrow$ shifts right

Poisson Regression Intuition (Simple)

Goal: Predict number of events **Model:** $\lambda = e^{X\beta}$

- Y = number of doctor visits (what we observe)
- λ = expected number of visits (what model predicts)

What is X ? (inputs)

- Age
- Income
- Health status

What is β ?

- How strongly each variable affects visits

Example:

- If $\beta_{\text{age}} = 0.1$
- Increasing age by 1 year multiplies visits by: $[e^{0.1} \approx 1.105]$

Interpretation:

- +1 in $X \Rightarrow$ multiply expected count (not add!)

Problem 1: Overdispersion

What Poisson assumes: [Mean = Variance]

Reality (real data): [Variance > Mean]

What does this mean? (simple words)

- Data is more "spread out" than expected
- Too many very small AND very large values

Example (doctor visits):

- Average visits = 2
- But some people go 0 times
- Some people go 10+ times

→ **Data is very uneven = overdispersion**

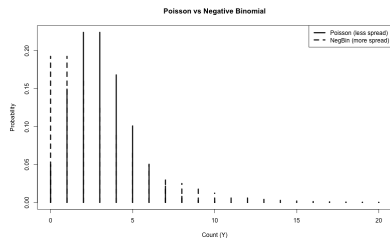
Problem:

- Poisson underestimates variability
- Leads to wrong conclusions

Solution:

- Use **Negative Binomial model**
- It allows extra variability

Negative Binomial Model (Simple)



More spread than Poisson (fatter tails)

Problem (Poisson fails):

- Data is too spread out
- Many zeros AND some very large values

Idea:

- Add extra randomness
- Allow variance to be bigger than mean

Model: [$Y \sim \text{NegBin}(\mu, \theta)$]

- μ = expected count (like Poisson)
- θ = extra variability

Key difference: [Variance > Mean]

When to use:

- Data is very uneven (overdispersion)

Negative Binomial Distribution (Intuition)

Simple idea:

- We repeat trials until we get v successes
- If success is rare $\rightarrow$ we need many trials $\rightarrow$ data is very spread out
- This extra spread is called **overdispersion**

Mathematical form:

$$\mathbb{P}(Y = y) = \binom{y-1}{v-1} p^v q^{y-v}, \quad k = v, v+1, \dots, \quad q = 1 - p.$$

$$\mathbb{E}(Y) = \frac{v}{p}$$

$$\text{Var}(Y) = \frac{vq}{p^2}$$

What parameters mean:

- p = probability of success (how likely success is)
- $q = 1 - p$ = probability of failure
- v = number of successes we want (target)
- Y = total number of trials needed

Important intuition:

- Small $p \rightarrow$ success is rare $\rightarrow Y$ becomes large
- Some observations are small, some very large $\rightarrow$ big variation

Why better than Poisson?

- Poisson: variance = mean (too restrictive)
- Negative Binomial: variance $>$ mean
- Handles real-world messy data better

Problem 2: Too Many Zeros

What is the problem?: Data has **a lot of zeros** More zeros than the model expects

Example (hospital visits):

- Many people go 0 times (perfectly healthy)
- Some people go many times

Why is this a problem?

- Poisson assumes zeros happen "naturally"
- But in reality, some people are **always zero**

Simple intuition:

- There are **two types of people**:
 - Type 1: never go (always 0)
 - Type 2: sometimes go (0,1,2,...)

Problem with Poisson:

- It mixes everyone together
- Cannot explain too many zeros

Solution:

- Use **Zero-Inflated models** model:
 - "always zero" group and "count" group

Zero-Inflated Poisson (ZIP): Math Made Simple

Idea: There are **two ways** to get a zero.

- Type 1: person is in the **always-zero group**
- Type 2: person is in the **count group**, and Poisson also gives 0

Step 1: Probability of being always zero [$P(Y_i = 0 \mid x_i, z_i) = \psi_i + (1 - \psi_i)e^{-\lambda_i}$]

Meaning:

- ψ_i = probability that observation i belongs to the always-zero group
- $(1 - \psi_i)$ = probability that observation i belongs to the count group
- $e^{-\lambda_i}$ = probability of zero from a Poisson model

Step 2: Probability of a positive count [$P(Y_i = y_i \mid x_i, z_i) = (1 - \psi_i) \frac{e^{-\lambda_i} \lambda_i^{y_i}}{y_i!}, \quad y_i = 1, 2, 3, \dots$]

Meaning:

- Positive counts can only come from the count group
- Then we use the usual Poisson probability

How do we model these two parts? [$\psi_i = F(z_i\gamma), \quad \lambda_i = e^{x_i\beta}$]

- $F(z_i\gamma)$ = logit/probit model for being always zero
- $e^{x_i\beta}$ = Poisson mean for the count group

ZIP Model: What the Math Really Says

Think of hospital visits:

- Some people will **never** go to hospital
- Other people may go 0, 1, 2, 3, ... times

So zero can happen in two different ways:

- Person is an **always-zero person**
- Person is a **normal count person**, but this time count happened to be 0

That is why ZIP is better than Poisson:

- Poisson has only one source of zeros
- ZIP has two sources of zeros
- So ZIP handles **too many zeros** better

Memory trick:

- ψ_i = chance of being **always zero**
- λ_i = average count for the **count group**

Hurdle Models

Idea: Two separate decisions:

- First: zero or not?
- Second: how many (if not zero)?

Step 1: Binary model

- Models whether $Y = 0$ or $Y > 0$
- Example: go to hospital or not

Step 2: Count model

- Only for positive values ($Y > 0$)
- Uses truncated Poisson or NegBin
- Zero is NOT allowed here
- Step 1: Does person visit doctor? (yes/no)
- Step 2: If yes $\rightarrow$ how many visits?

Key intuition:

- You must cross a "hurdle" to leave zero
- After that $\rightarrow$ normal count process
- Zeros and positive counts come from different behavior

ZIP vs Hurdle Models (Very Simple)

Zero-Inflated Model (ZIP)

Idea:

- Two types of people:
 - Always zero
 - Count group

Zeros come from TWO sources:

- Always-zero group
- Poisson model

Key feature:

- Count model can produce zeros

Example:

- Some people never go to hospital
- Others sometimes go (including 0)

BIG DIFFERENCE (remember this!):

- ZIP: zeros come from TWO processes
- Hurdle: zeros come from ONE process only

Hurdle Model

Idea:

- Must cross a "hurdle" to leave zero

Zeros come from ONE source:

- Binary model only

Key feature:

- Count model has NO zeros

Example:

- First: go to hospital or not
- Then: how many visits (only positive)

Model Selection Guide (More Complete)

1. Start with Poisson

- Assumes observations are independent
- Basic choice for count data

2. If data is overdispersed ; ($\text{Var}(Y) > E(Y)$)

- **Quasi-Poisson**: Poisson mean, but robust variance
- **Negative Binomial**: directly models overdispersion

3. If data has too many zeros ; (zero inflation)

- **Zero-Inflated Poisson (ZIP)**
- **Zero-Inflated Negative Binomial (ZINB)**
- **Hurdle models**
- **Finite mixture models** and more

Memory trick:

- Simple count data then Poisson
- Too much spread then NegBin / Quasi-Poisson
- Too many zeros then ZIP / ZINB / Hurdle

Model Comparison Table

Model	When to use	Example	Intuition
Poisson	Mean $\approx$ variance; independent events in fixed interval	Calls per hour	Random arrivals with constant rate
Negative Binomial	Variance $>$ mean (overdispersion)	Accidents	Poisson with extra variability
ZIP	Excess zeros beyond Poisson	Hospital visits (many non-users)	Zeros come from two sources: structural zeros + count model (which can also produce zeros)
ZINB	Excess zeros + overdispersion	Insurance claims	Same as ZIP but with flexible variance
Hurdle	Zero vs positive outcomes are separate processes	Gym visits per month	First hurdle: zero vs nonzero; positives use truncated model (no zeros allowed)

Thank you